AI101 MINI-PROJECT

Human Activity Recognition Using Smartphone Sensors

Classifying 6 everyday activities from accelerometer and gyroscope data

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Repository: github.com/CHADMONK07/Human-Activity-Recognition-Using-Smartphone

Problem and data

What and why

Phones carry motion sensors. Turning raw movement into an activity label powers health tracking, fitness apps and fall or safety alerts.

Goal

Train and compare three classifiers, then pick the best for a prediction pipeline.

Dataset

UCI Human Activity Recognition Using Smartphones: accelerometer and gyroscope signals with 561 engineered features (time and frequency domain).

561
FEATURES

6
ACTIVITY CLASSES

2,947
TEST SAMPLES

Classes (label 0 to 5)

- 0 Walking
- 1 Upstairs
- 2 Downstairs
- 3 Sitting
- 4 Standing
- 5 Laying

Approach



Logistic Regression

Linear baseline

sklearn defaults

Random Forest

Ensemble of trees

n_estimators=100, random_state=42

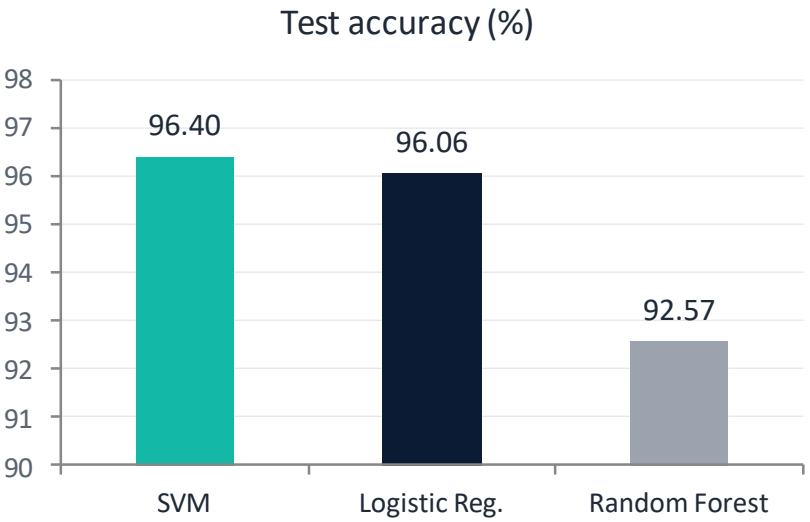
SVM

Max-margin classifier

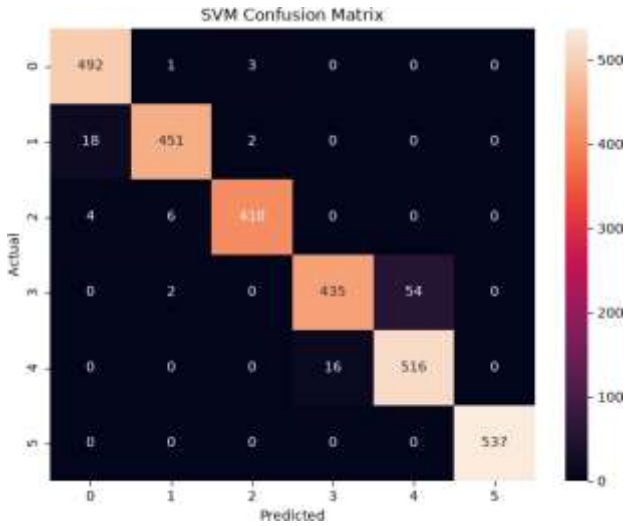
kernel = linear

Libraries: **Python, pandas, NumPy, scikit-learn, Matplotlib, Seaborn, joblib**. Same train and test data for every model; judged by accuracy, classification report and confusion matrix.

Results: SVM wins at 96.40%



Value axis starts at 90% to show the gap.



SVM confusion matrix (test set)

2,841 / 2,947

SVM(BEST MODEL FOR THIS DATA SET)

100%

LAYING RECALL

70 of 106

SVM ERRORS: SITTING VS STANDING

Learnings

What worked

- Linear models suit 561 engineered features
- Same split for all models gave a fair comparison
- Laying is 100% correct in every model

What didn't

- Sitting vs Standing stays confused (~70 errors)
- Random Forest trailed by about 4 points
- Next: tune hyperparameters, try LSTM / CNN

RESULTS

Model	Algorithm Type	Accuracy
Logistic Regression	Linear Classification Model	96.06%
Random Forest	Ensemble Tree-Based Model	92.57%
Support Vector Machine (SVM)	Margin-Based Classification Model	96.40%

AI-use disclosure : Claude (Anthropic) for PPT and finalising code
CHATGPT(OpenAI) for code , repository and git management